Particle Model of Gases

Exam code: 0625 & 0972

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Motion of particles in a gas

- In the particle model, molecules in a gas are in constant **random** motion at high speeds
- Random motion means that the molecules are travelling in no specific path and undergo sudden changes in their motion if they collide:
  - With the walls of its container
  - With other molecules
- Pressure in a gas is caused by the collisions of particles with the walls of the container
- When the particles travel **faster** (e.g. at a higher temperature), they collide with walls **more frequently**
  - This means the gas exerts a **greater pressure**

Help me understand

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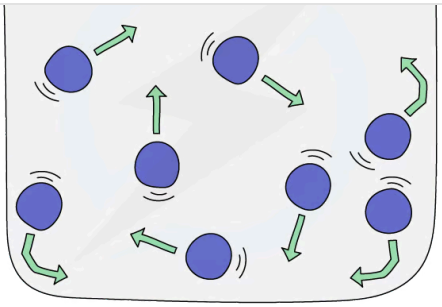
Got it



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Random motion of gas molecules in a container. Collisions with container walls creates pressure in the particle model

Pressure & force of particles in a gas

Extended tier only

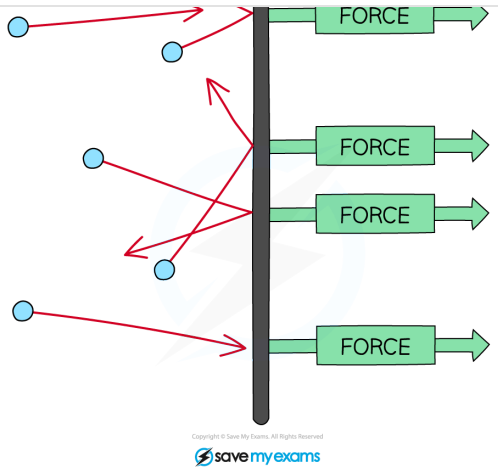
- Gases fill their container
- The pressure is defined as the **force per unit area**

$$p = \frac{F}{A}$$

- Where:
  - p = pressure in pascals Pa
  - F = force in newtons N
  - A = area in metres-squared m^2

- This equation and the particle model can be used to explain how particles exert pressure
- As the gas particles move about randomly they collide with the walls of their containers
- These collisions produce **force** at right angles to the wall of the gas container (or any surface)
  - Pressure is force per unit area, so the force of these collisions exerts a **pressure**

- When the particles move faster, they have **more frequent collisions** with the container walls and these exert a **greater force**
  - If the **force** exerted per unit area is greater and more frequent, the **pressure** exerted is also greater



Gas molecules bouncing off the walls of a container exert a force perpendicular to the surface

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